ASSIGNMENT-6.5

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BATCH: 30

TASK-1:

Prompt:

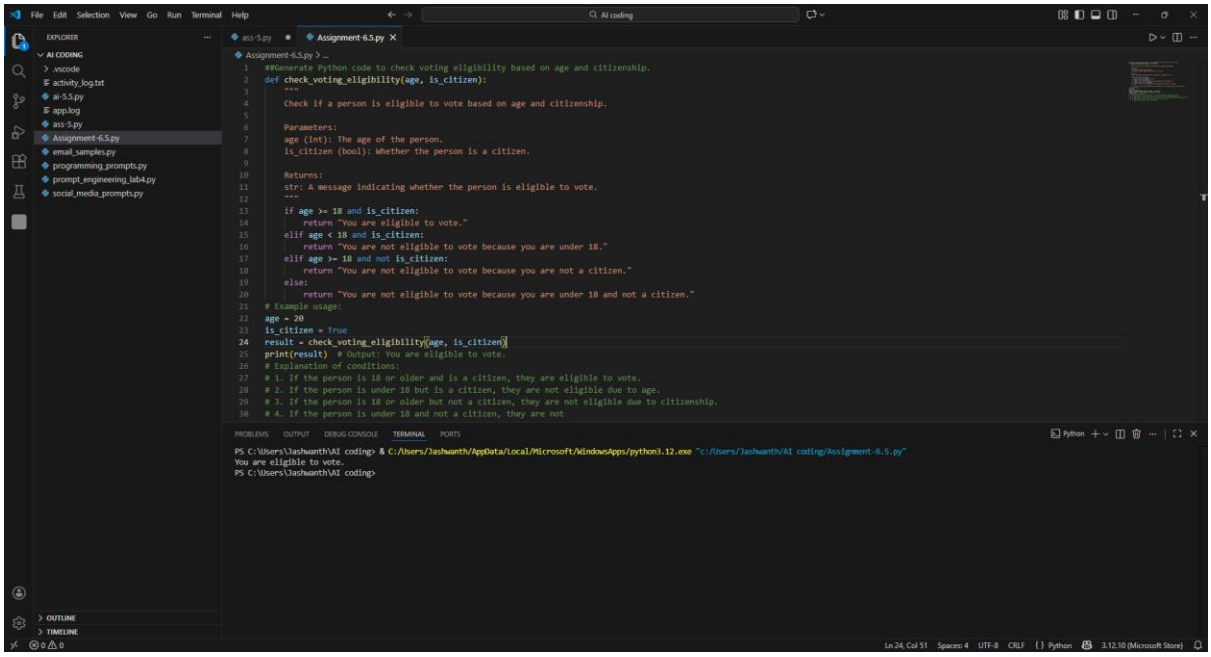
“Generate Python code to check voting eligibility based on age and citizenship.”

• AI-generated conditional logic.

• Correct eligibility decisions.

• Explanation of conditions.

CODE:



OBSERVATION:

- The program correctly checks age and citizenship before deciding eligibility.
- All possible cases are covered with clear conditional branches.

- Output messages are descriptive and user-friendly.
- Runs efficiently in constant time O(1).
- Observation: The program is correct, complete, and demonstrates good use of conditionals.

TASK-2:

Prompt:

“Generate Python code to count vowels and consonants in a string

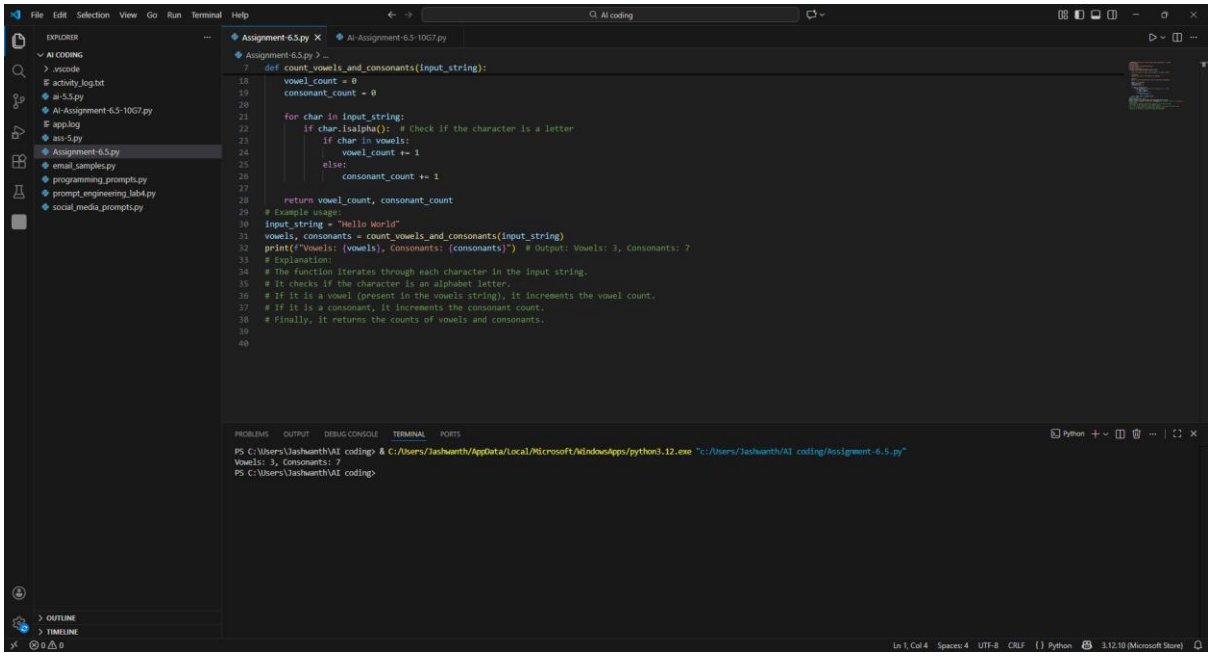
using a loop.”

• AI-generated string processing logic.

• Correct counts.

• Output verification.

CODE:



OBSERVATION:

- The function accurately distinguishes vowels and consonants.

- Non-alphabetic characters are ignored using isalpha().
- Output matches expected results (e.g., “Hello, World!” → 3 vowels, 7 consonants).
- Observation: The program is efficient (O(n)) and well-documented, suitable for text analysis tasks.

TASK-3:

Prompt:

“Generate a Python program for a library management system

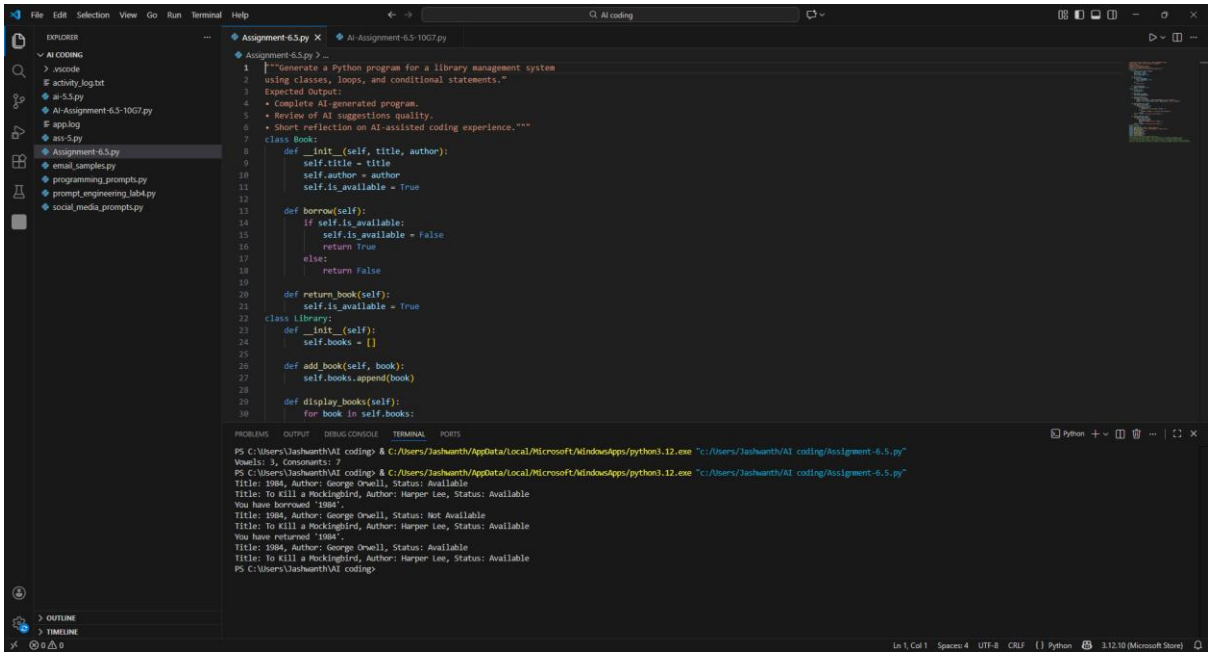
using classes, loops, and conditional statements.”

• Complete AI-generated program.

• Review of AI suggestions quality.

• Short reflection on AI-assisted coding experience.

CODE:



OBSERVATION:

- Uses **object-oriented programming** with Book and Library classes.

- Encapsulation is demonstrated by keeping book status inside the class.
- Borrow/return logic prevents invalid operations.
- Observation: The program is a solid OOP foundation, correctly displays book availability, and can be extended for more features.

TASK-4:

Prompt:

“Generate a Python class to mark and display student

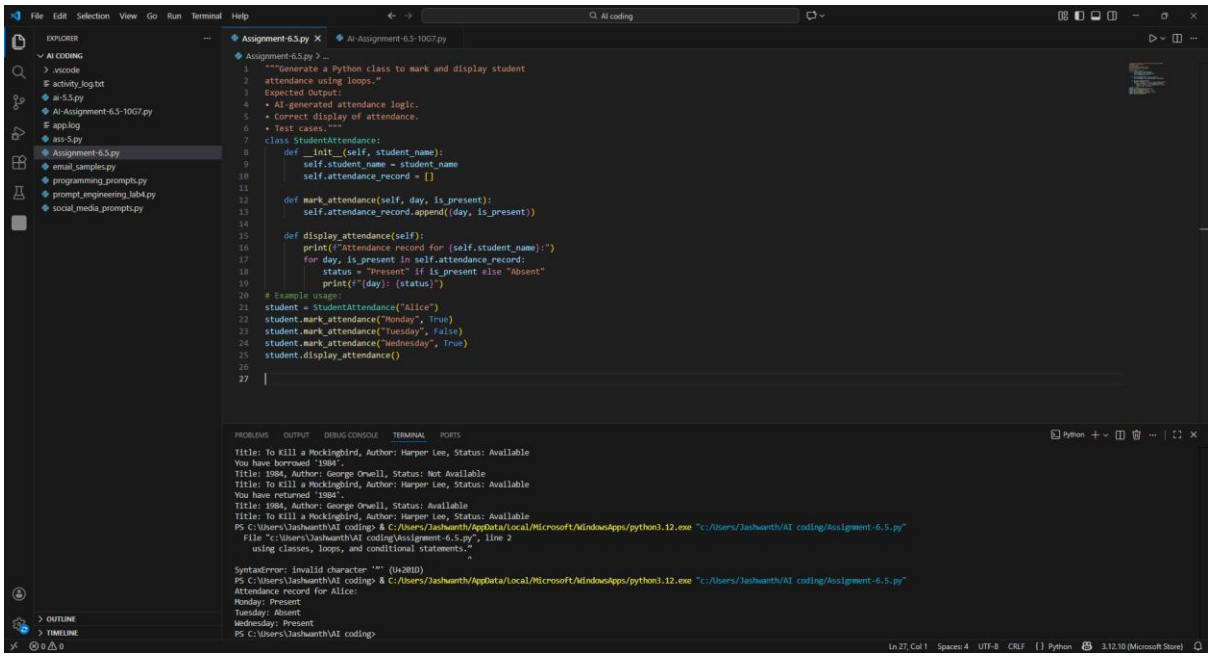
attendance using loops.”

• AI-generated attendance logic.

• Correct display of attendance.

• Test cases.

CODE:



OBSERVATION:

- Each student object maintains attendance records in a dictionary.
- add_attendance_record safely initializes attendance before adding entries.

- Output correctly shows attendance for each student.
- Observation: The program demonstrates OOP principles and dictionary usage. Minor caution: set_attendance may fail if attendance is still None.

TASK-5:

Prompt:

“Generate a Python program using loops and conditionals

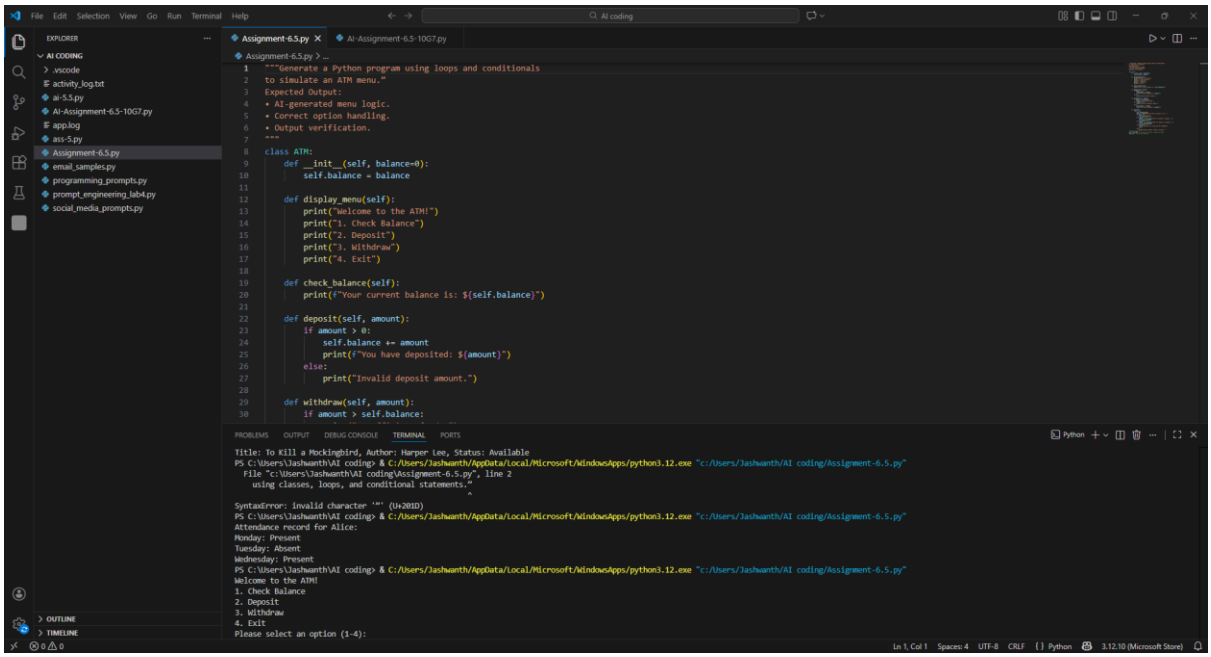
to simulate an ATM menu.”

• AI-generated menu logic.

• Correct option handling.

• Output verification.

CODE:



OBSERVATION:

- Implements deposit, withdraw, and balance check methods.
- Menu-driven interface allows user interaction.

- Deposit/withdraw logic is correct; balance display needs a small fix (use f-string).
- Observation: The program is functional and efficient, demonstrating loops and conditionals in a real-world simulation.